

Project Documentation

Project title: **SCRIPT-CONTROLLED ACL – RESTRICT RECORD ACCESS BASED ON FIELD VALUE**

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ABSTRACT

Access Control Lists (ACLs) are an important security mechanism in ServiceNow used to control who can access, create, modify, or delete records. This project focuses on implementing a **script-controlled ACL mechanism to restrict record access based on a field value**. The project is developed using the ServiceNow Developer Instance. A student/institution details table is created with fields such as Student Roll Number, Student Name, Faculty Name, Branch, Phone Number, and Description. Different custom roles are created to provide different levels of access to the records. The main objective is to ensure that users do not receive unrestricted access to all records. For example, the Read ACL is configured with the `bb1` role and a condition that restricts access to records where the **Branch is EEE**. A scripted ACL is also used to control access based on the user's role. Separate ACLs are created for Write, Create, and Delete operations using the `bb2` and `bb4` roles. This project demonstrates how ServiceNow ACLs, roles, conditions, and scripts can be combined to implement **role-based and field-based record security**.

1.Introduction

Access control is an important part of application security. In an application, different users may require different levels of access.

For example:

- Administrators may have full access.
- Managers may have limited access.
- Normal users may have read-only access.

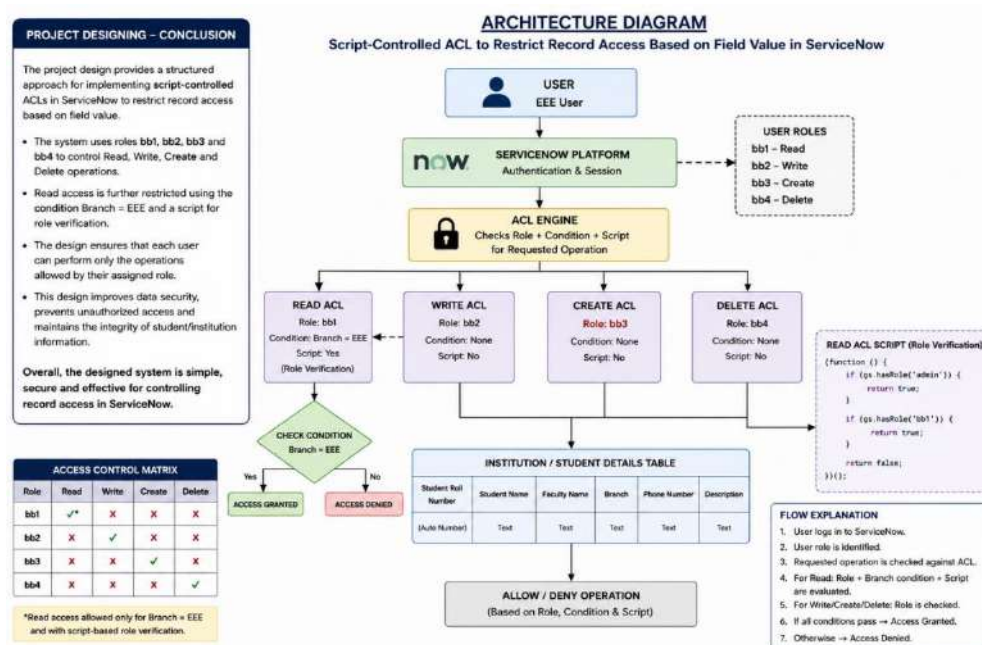
The proposed system uses **ACLs and roles** to implement these permissions. The system allows administrators to create roles, assign permissions, configure ACL rules, and remove unnecessary roles when required.

1.1. Objectives:

The main objectives of the project are:

- To provide secure access to application resources.
- To create and manage user roles.
- To create and configure Access Control Lists.
- To control read access to resources.
- To assign appropriate permissions to users.
- To prevent unauthorized access.
- To remove unnecessary roles when required

2. Architecture:



3. System Requirements

3.1. Software Requirements:

- **Operating System:** Windows / Linux
- **Platform:** ServiceNow
- **Web Browser:** Google Chrome / Microsoft Edge
- **Internet Connection:** Required
- **Database:** ServiceNow database
- **Development Environment:** ServiceNow Application Navigator and ACL configuration

3.2. Hardware Requirements:

- **Processor:** Intel Core i3 or above
- **RAM:** Minimum 4 GB
- **Storage:** Minimum 10 GB free space
- **Input:** Keyboard and Mouse
- **Display:** Standard monitor
- **Internet:** Stable internet connection.

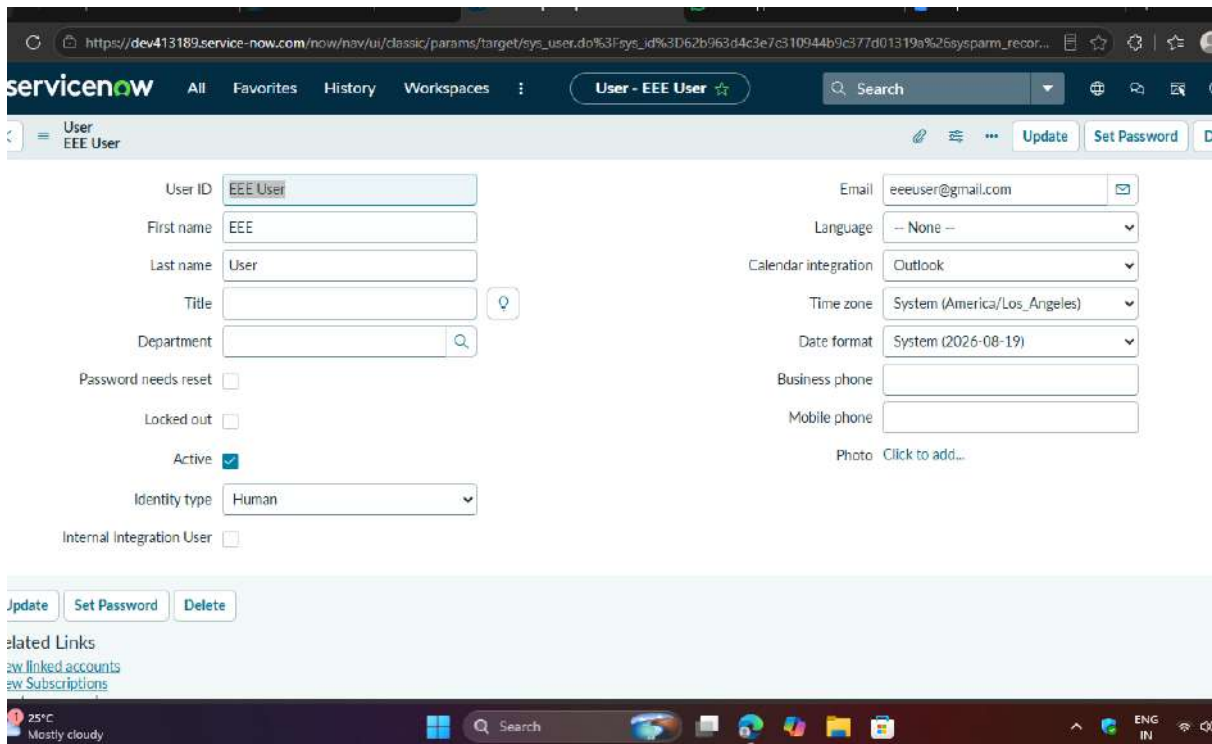
4. Implementation:

The project is implemented using the role and ACL management features of ServiceNow.

Milestone-1 Creation of Users and Roles:

User Creation:

1. **Log in** to the ServiceNow instance with **admin access**.
2. **Create a test user:**
 - Navigate to **User Administration** → **Users**
 - Click **New**
 - **User ID:** EEE User
 - **First name:** EEE
 - **Last name:** User
 - **Email:** eeeuser@gmail.com
 - Save the record

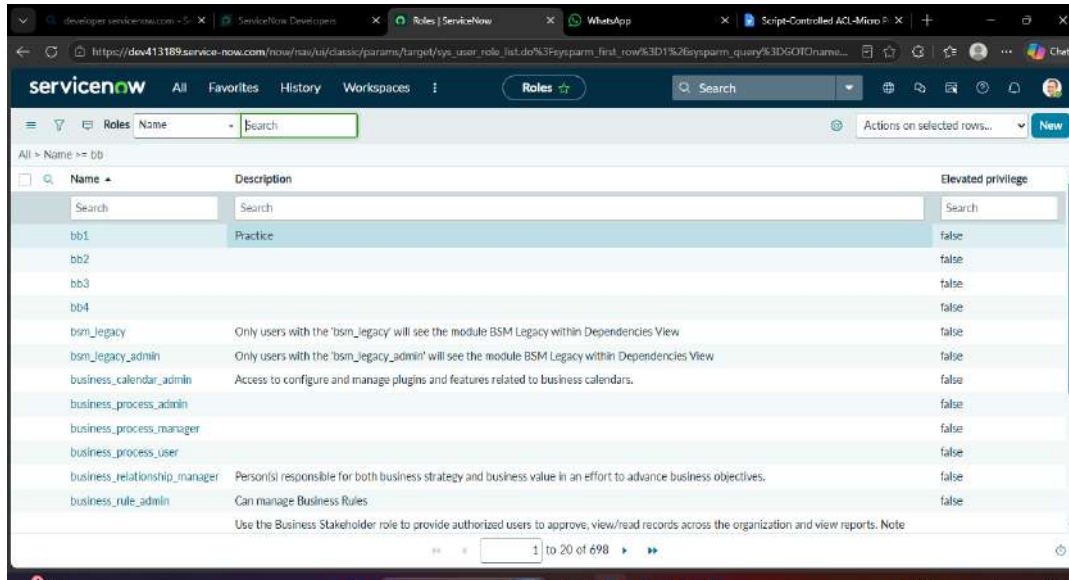


The screenshot shows the ServiceNow 'User - EEE User' form. The form is divided into two main sections. The left section contains fields for 'User ID' (filled with 'EEE User'), 'First name' (filled with 'EEE'), 'Last name' (filled with 'User'), 'Title' (empty), 'Department' (empty), 'Password needs reset' (checkbox), 'Locked out' (checkbox), 'Active' (checkbox, checked), 'Identity type' (dropdown, 'Human'), and 'Internal Integration User' (checkbox). The right section contains fields for 'Email' (filled with 'eeeuser@gmail.com'), 'Language' (dropdown, 'None'), 'Calendar integration' (dropdown, 'Outlook'), 'Time zone' (dropdown, 'System (America/Los Angeles)'), 'Date format' (dropdown, 'System (2026-08-19)'), 'Business phone' (empty), 'Mobile phone' (empty), and 'Photo' (button 'Click to add...'). At the bottom of the form are buttons for 'Update', 'Set Password', and 'Delete'. Below the buttons are 'Related Links' for 'Linked accounts' and 'Subscriptions'. The bottom of the screenshot shows a Windows taskbar with a search bar and various application icons.

Role Creation:

- Navigate to **User Administration** → **Roles**
- Click **New**
- **Name:** Enter role name as bb1
- **Description:** Brief role purpose

- Click **Submit**
- Create other roles also like **bb2,bb3,bb4.(follow the same procedure)**
- Assign all the roles to the **EEE User**



Milestone-2: Tables Creation:

Tables Creation:

1. In the **Application Navigator**, search for **System Definition → Tables**.
2. Click **New** and create a custom table with the following details:
 - **Label:** Institution Details
 - **Name:** u_institution_details
 - **Extends for :** false
3. .Click **Submit**.
4. Add the following fields to the **u_institution_details** table:
 - **Student Roll Number**– Auto Number
 - **Student Name** – Reference- User
 - **Faculty Name** – Reference- User
 - **Branch** – Choice (ECE, EEE, CSE)
 - **Email**– String
 - **Phone Number**– String

- **Description**– Multi String

5. Save the table

6. Create multiple records in the **Student Records** table with different branch values (ECE, EEE, CSE).

ServiceNow recommends creating custom tables in scoped applications. To learn more about creating scoped applications, click [here](#).

A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More Info](#)

* Label:

* Name:

Extends table:

Application:

Create module: ☒

Create mobile module: ☒

Add module to menu:

New menu name:

Remote Table: ☐

Columns | Controls | Application Access

Table Columns: Search

Dictionary Entries

Table Institution Details				
	Column label	Type	Reference	Max length
	Updated	Date/Time	(empty)	40
✖	Student Name	Reference	User	32
✖	Student Roll Number	String	(empty)	40
	Created by	String	(empty)	40
	Sys ID	Sys ID (GUID)	(empty)	32
✖	Faculty Name	Reference	User	32
✖	Phone Number	String	(empty)	40
✖	Description	String	(empty)	40
	Created	Date/Time	(empty)	40
✖	Email	String	(empty)	40
	Updated by	String	(empty)	40
	Updates	Integer	(empty)	40
✖	Branch	Choice	(empty)	40
+	Insert a new row...			

Delete
Update
Delete All Records

Institution Details							
	Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
	ECE	N there	Abraham@gmail.com	Abraham Lincoln	8099775510	Abel Tuter	STU100
	(empty)			(empty)		(empty)	
	EEE	656	Aileen@gmail.com	Aileen Mottern	1234567891	Adela Cervantsz	STU200
	CSE	NC	Amos@gmail.com	Amos Linnan	5464745510	Alissa Mountjoy	STU300

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Milestone-3: Creation of Access Control List-READ

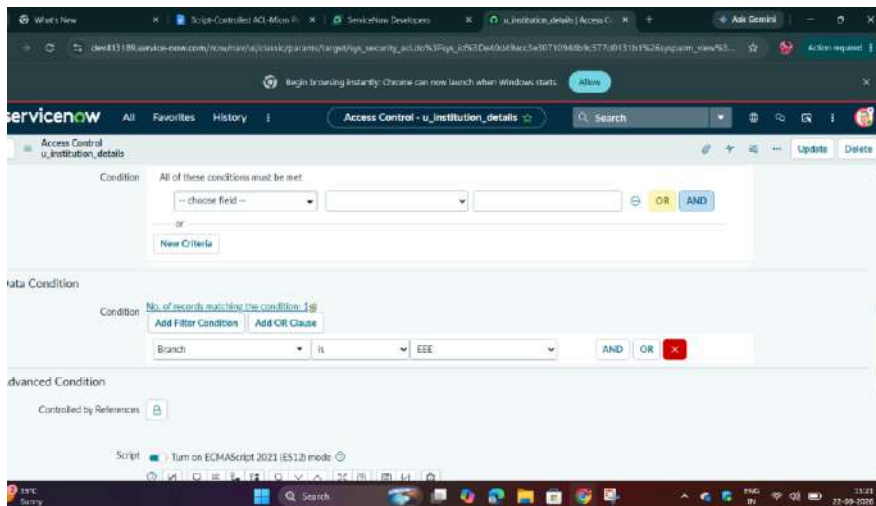
Creation of Access Control

1. **Click New to create a new ACL (Write).**
2. Elevate “Security_admin” role
3. . In the **Application Navigator**, search for **System Security → Access Control**
4. Enter the following details:
 - **Type:** record
 - **Operation:** read
 - **Name:** u_institution_details
 - **Active:** true
 - **Advanced:** true
5. In the **Requires role** related list, add:
 - bb1 (custom role)
6. Add Data condition as **Branch is EEE**
7. In the **Script** field, add the following script:

Lists(Read ACL):

```
(function () {  
  
    // Allow admin users full access  
    If (gs.hasRole('admin')) {  
        return true;  
    }  
    // Allow only EEE branch users to see EEE records  
    if (gs.hasRole('bb1')){  
        return true;  
    }  
    // Deny access for all others  
    return false;  
})();
```

8. Click **Submit** to save the ACL.



Milestone-4: Creation of Access Control List-CREATE:

Creation of Access Control Lists(Create ACL)

1. **Click New to create a new ACL (Create)**
2. Enter the following details:
 - **Type:** record
 - **Operation:** create
 - **Name:** u_institution_details
 - **Active:** true
3. In the **Requires role** related list, add:
 - bb2 (custom role)
4. No need to add any Data condition
5. Save the ACL Record

Milestone-5: Creation of Access Control List-WRITE:

Creation of Access Control Lists(Create ACL):

1. **Click New to create a new ACL(Create)**
2. Enter the following details:
 - **Type:** record

- **Operation:** create
 - **Name:** u_institution_details
 - **Active:** true
3. In the **Requires role** related list, add:
 - bb2 (custom role)
 4. No need to add any Data condition
 5. Save the ACL Record

Milestone-6: Creation of Access Control List-DELETE:

Creation of Access Control Lists(Delete ACL)

1. **Click New to create a new ACL(Delete)**
2. Enter the following details:
 - **Type:** record
 - **Operation:** Delete
 - **Name:** u_institution_details
 - **Active:** true
3. In the **Requires role** related list, add:
 - bb4 (custom role)
4. No need to add any Data condition

Save the ACL Record

The screenshot shows the ServiceNow 'Access Controls' page. The breadcrumb trail is 'All > Name > u_institution_details'. The table below lists various ACLs with columns for Name, Decision Type, Operation, Type, Active status, Updated by, and Updated date.

Name	Decision Type	Operation	Type	Active	Updated by	Updated
u_institution_details	Allow If	write	record	true	admin	2026-09-22 01:03:35
u_institution_details	Allow If	create	record	true	admin	2026-09-22 00:58:59
u_institution_details	Allow If	read	record	true	admin	2026-09-21 17:54:52
u_institution_details	Allow If	create	record	true	admin	2026-09-21 17:54:52
u_institution_details	Allow If	delete	record	true	admin	2026-09-22 01:07:58
u_institution_details	Allow If	delete	record	true	admin	2026-09-21 17:54:52
u_institution_details	Allow If	read	record	true	admin	2026-09-22 00:49:55
u_institution_details	Allow If	write	record	true	admin	2026-09-21 17:54:52
u_licensing_exclusion	Allow If	create	record	true	admin	2017-08-15 13:09:53
u_licensing_exclusion	Allow If	delete	record	true	admin	2017-08-15 13:10:11
VA Channel Integration ACL	Allow If	execute	REST_Endpoint	true	admin	2023-04-05 00:28:13

The screenshot shows the ServiceNow 'Institution Details' page. The breadcrumb trail is 'All > Student Roll Number'. The table below lists student records with columns for Student Roll Number, Student Name, Branch, Faculty Name, Description, Email, and Phone Number.

Student Roll Number	Student Name	Branch	Faculty Name	Description	Email	Phone Number
STU100	Abel Tuter	ECE	Abraham Lincoln	N there	Abraham@gmail.com	8099775510
(empty)	(empty)		(empty)			
STU200	Adela Cervantsz	EEE	Alien Mottem	656	Alien@gmail.com	1234567891
STU300	Alissa Mountjoy	CSE	Amos Linnan	NC	Amos@gmail.com	5464745510

5.Conclusion

The Access Control List Management System provides a secure and organized method for controlling access to application resources. By using roles, permissions, and ACLs, the system allows authorized users to access required resources while restricting unauthorized users. The implementation improves application security and makes user access easier to manage.

